

HW4

RAJ SHAH

December 2025

Problem 1: EM Algorithm for a Gaussian Mixture Model

Using the four data points provided below, perform a single iteration of the Expectation-Maximization (EM) algorithm for a Gaussian Mixture Model (GMM):

$$f_X(x) = \pi_0 \cdot \mathcal{N}(x; \mu_0, \sigma_0^2) + \pi_1 \cdot \mathcal{N}(x; \mu_1, \sigma_1^2).$$

The data points are:

data num	x
d_1	2
d_2	1
d_3	-1
d_4	-2

We are given the initial parameters:

$$\pi_0^{(t)} = \pi_1^{(t)} = \frac{1}{2}, \quad \mu_0^{(t)} = -1, \quad \mu_1^{(t)} = 1, \quad \sigma_0^{2(t)} = \sigma_1^{2(t)} = 1.$$

Denote the four observations as

$$x_1 = 2, \quad x_2 = 1, \quad x_3 = -1, \quad x_4 = -2.$$

1.1 Log-Likelihood Computation

The model density for a single observation x is

$$f(x) = \pi_0 \mathcal{N}(x; \mu_0, \sigma_0^2) + \pi_1 \mathcal{N}(x; \mu_1, \sigma_1^2),$$

with

$$\pi_0 = \pi_1 = \frac{1}{2}, \quad \mu_0 = -1, \quad \mu_1 = 1, \quad \sigma_0^2 = \sigma_1^2 = 1.$$

For a 1-D Gaussian with variance 1,

$$\mathcal{N}(x; \mu, 1) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2}\right).$$

We first compute $f(x)$ for each x_i .

For $x = 2$:

$$\mathcal{N}(2; -1, 1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(2-(-1))^2}{2}} = \frac{1}{\sqrt{2\pi}} e^{-\frac{9}{2}},$$

$$\mathcal{N}(2; 1, 1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(2-1)^2}{2}} = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}}.$$

Hence,

$$f(2) = \frac{1}{2} \mathcal{N}(2; -1, 1) + \frac{1}{2} \mathcal{N}(2; 1, 1) = \frac{1}{2\sqrt{2\pi}} \left(e^{-9/2} + e^{-1/2} \right).$$

For $x = 1$:

$$\mathcal{N}(1; -1, 1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(1+1)^2}{2}} = \frac{1}{\sqrt{2\pi}} e^{-2},$$

$$\mathcal{N}(1; 1, 1) = \frac{1}{\sqrt{2\pi}} e^0 = \frac{1}{\sqrt{2\pi}}.$$

Thus,

$$f(1) = \frac{1}{2} \mathcal{N}(1; -1, 1) + \frac{1}{2} \mathcal{N}(1; 1, 1) = \frac{1}{2\sqrt{2\pi}} (e^{-2} + 1).$$

For $x = -1$: By symmetry,

$$f(-1) = \frac{1}{2\sqrt{2\pi}} (e^{-2} + 1).$$

For $x = -2$: Again by symmetry with $x = 2$,

$$f(-2) = \frac{1}{2\sqrt{2\pi}} \left(e^{-9/2} + e^{-1/2} \right).$$

The log-likelihood for the four independent points is

$$\ell = \sum_{i=1}^4 \log f(x_i) = 2 \log \left[\frac{1}{2\sqrt{2\pi}} (e^{-9/2} + e^{-1/2}) \right] + 2 \log \left[\frac{1}{2\sqrt{2\pi}} (e^{-2} + 1) \right].$$

Evaluating numerically (natural logarithm),

$$\ell \approx -7.16.$$

$$\boxed{\ell \approx -7.16}$$

1.2 E-step: Responsibilities γ_{nk}

In the E-step, we compute the posterior responsibility that each mixture component $k \in \{0, 1\}$ has for each data point x_n :

$$\gamma_{n0} = \frac{\pi_0 \mathcal{N}(x_n; \mu_0, \sigma^2)}{\pi_0 \mathcal{N}(x_n; \mu_0, \sigma^2) + \pi_1 \mathcal{N}(x_n; \mu_1, \sigma^2)}, \quad \gamma_{n1} = 1 - \gamma_{n0},$$

with

$$\pi_0 = \pi_1 = \frac{1}{2}, \quad \mu_0 = -1, \quad \mu_1 = 1, \quad \sigma^2 = 1.$$

Since both components have the same variance and the same prefactor, the Gaussian normalizing constants cancel in the ratio, giving

$$\gamma_{n0} = \frac{\exp\left(-\frac{(x_n+1)^2}{2}\right)}{\exp\left(-\frac{(x_n+1)^2}{2}\right) + \exp\left(-\frac{(x_n-1)^2}{2}\right)}.$$

We now compute these values for each data point.

For $x_1 = 2$:

$$\gamma_{10} = \frac{e^{-\frac{(2+1)^2}{2}}}{e^{-\frac{(2+1)^2}{2}} + e^{-\frac{(2-1)^2}{2}}} = \frac{e^{-4.5}}{e^{-4.5} + e^{-0.5}} \approx \frac{0.0111}{0.0111 + 0.6065} \approx 0.0180,$$

$$\gamma_{11} = 1 - \gamma_{10} \approx 0.9820.$$

For $x_2 = 1$:

$$\gamma_{20} = \frac{e^{-\frac{(1+1)^2}{2}}}{e^{-\frac{(1+1)^2}{2}} + e^{-\frac{(1-1)^2}{2}}} = \frac{e^{-2}}{e^{-2} + 1} \approx \frac{0.1353}{0.1353 + 1} \approx 0.1192,$$

$$\gamma_{21} = 1 - \gamma_{20} \approx 0.8808.$$

For $x_3 = -1$:

$$\gamma_{30} = \frac{e^{-\frac{(-1+1)^2}{2}}}{e^{-\frac{(-1+1)^2}{2}} + e^{-\frac{(-1-1)^2}{2}}} = \frac{1}{1 + e^{-2}} \approx \frac{1}{1 + 0.1353} \approx 0.8808,$$

$$\gamma_{31} = 1 - \gamma_{30} \approx 0.1192.$$

For $x_4 = -2$:

$$\gamma_{40} = \frac{e^{-\frac{(-2+1)^2}{2}}}{e^{-\frac{(-2+1)^2}{2}} + e^{-\frac{(-2-1)^2}{2}}} = \frac{e^{-0.5}}{e^{-0.5} + e^{-4.5}} \approx \frac{0.6065}{0.6065 + 0.0111} \approx 0.9820,$$

$$\gamma_{41} = 1 - \gamma_{40} \approx 0.0180.$$

n	x_n	γ_{n0}	γ_{n1}
1	2	0.0180	0.9820
2	1	0.1192	0.8808
3	-1	0.8808	0.1192
4	-2	0.9820	0.0180

1.3 M-step: Updated Parameters

In the M-step, we update the parameters using the responsibilities γ_{nk} .

1.3.1 Effective Counts

Define the effective number of points assigned to each component:

$$N_0 = \sum_{n=1}^4 \gamma_{n0}, \quad N_1 = \sum_{n=1}^4 \gamma_{n1}.$$

Using the values from the E-step:

$$N_0 = 0.0180 + 0.1192 + 0.8808 + 0.9820 = 2.0000,$$

$$N_1 = 0.9820 + 0.8808 + 0.1192 + 0.0180 = 2.0000.$$

1.3.2 Updated Mixing Weights

The updated mixture weights are

$$\pi_0^{(t+1)} = \frac{N_0}{4} = \frac{2}{4} = 0.5, \quad \pi_1^{(t+1)} = \frac{N_1}{4} = 0.5.$$

1.3.3 Updated Means

The updated means are given by

$$\mu_0^{(t+1)} = \frac{1}{N_0} \sum_{n=1}^4 \gamma_{n0} x_n, \quad \mu_1^{(t+1)} = \frac{1}{N_1} \sum_{n=1}^4 \gamma_{n1} x_n.$$

For component 0:

$$\sum_{n=1}^4 \gamma_{n0} x_n = 0.0180(2) + 0.1192(1) + 0.8808(-1) + 0.9820(-2),$$

$$= 0.036 + 0.1192 - 0.8808 - 1.964 = -2.6896,$$

$$\mu_0^{(t+1)} = \frac{-2.6896}{2} = -1.3448.$$

For component 1:

$$\sum_{n=1}^4 \gamma_{n1} x_n = 0.9820(2) + 0.8808(1) + 0.1192(-1) + 0.0180(-2),$$

$$= 1.964 + 0.8808 - 0.1192 - 0.036 = 2.6896,$$

$$\mu_1^{(t+1)} = \frac{2.6896}{2} = 1.3448.$$

1.3.4 Updated Variances

For each component $k \in \{0, 1\}$, the updated variance is

$$\sigma_k^{2(t+1)} = \frac{1}{N_k} \sum_{n=1}^4 \gamma_{nk} (x_n - \mu_k^{(t+1)})^2.$$

Component 0: Using $\mu_0^{(t+1)} = -1.3448$, we compute the squared deviations:

$$(2 - (-1.3448))^2 = 3.3448^2 \approx 11.189,$$

$$(1 - (-1.3448))^2 = 2.3448^2 \approx 5.498,$$

$$(-1 - (-1.3448))^2 = 0.3448^2 \approx 0.119,$$

$$(-2 - (-1.3448))^2 = (-0.6552)^2 \approx 0.429.$$

Weighting by γ_{n0} :

$$\sum_{n=1}^4 \gamma_{n0} (x_n - \mu_0^{(t+1)})^2 \approx 0.0180(11.189) + 0.1192(5.498) + 0.8808(0.119) + 0.9820(0.429)$$

$$\approx 0.2014 + 0.6553 + 0.1047 + 0.4215 = 1.3829.$$

Thus,

$$\sigma_0^{2(t+1)} = \frac{1.3829}{2} \approx 0.6915.$$

Component 1: By symmetry with component 0 (since means are opposite and responsibilities mirror),

$$\sigma_1^{2(t+1)} \approx 0.6915.$$

$\pi_0^{(t+1)} = 0.5,$ $\pi_1^{(t+1)} = 0.5,$ $\mu_0^{(t+1)} = -1.3448,$ $\mu_1^{(t+1)} = 1.3448,$ $\sigma_0^{2(t+1)} \approx 0.6915,$ $\sigma_1^{2(t+1)} \approx 0.6915.$

1.4 Log-Likelihood Using the Updated Parameters

Using the parameters obtained from the M-step, we now compute the updated log-likelihood and verify that the EM algorithm increases it.

The updated parameters from part 1.3 are

$$\pi_0^{(t+1)} = \pi_1^{(t+1)} = 0.5, \quad \mu_0^{(t+1)} = -1.3448, \quad \mu_1^{(t+1)} = 1.3448,$$

$$\sigma_0^{2(t+1)} = \sigma_1^{2(t+1)} = 0.6915.$$

The Gaussian density with mean μ and variance σ^2 is

$$\mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right].$$

Since $\sigma^2 = 0.6915$, the normalizing constant becomes

$$\frac{1}{\sqrt{2\pi\sigma^2}} = \frac{1}{\sqrt{2\pi(0.6915)}} = \frac{1}{2.084} = 0.4798.$$

The mixture density after the M-step is

$$f(x) = 0.5\mathcal{N}(x; \mu_0^{(t+1)}, \sigma^2) + 0.5\mathcal{N}(x; \mu_1^{(t+1)}, \sigma^2).$$

We evaluate $f(x_n)$ for each data point.

Likelihood for $x_1 = 2$

Compute deviations:

$$2 - \mu_0^{(t+1)} = 3.3448, \quad 2 - \mu_1^{(t+1)} = 0.6552.$$

Exponent terms:

$$\frac{(3.3448)^2}{2\sigma^2} = \frac{11.189}{1.383} = 8.089, \quad \frac{(0.6552)^2}{2\sigma^2} = \frac{0.4293}{1.383} = 0.3106.$$

Thus,

$$\mathcal{N}(2; \mu_0, \sigma^2) = 0.4798e^{-8.089} = 0.000150, \quad \mathcal{N}(2; \mu_1, \sigma^2) = 0.4798e^{-0.3106} = 0.3544.$$

Mixture:

$$f(2) = 0.5(0.000150 + 0.3544) = 0.17725, \quad \log f(2) = -1.729.$$

Likelihood for $x_2 = 1$

$$1 - \mu_0^{(t+1)} = 2.3448, \quad 1 - \mu_1^{(t+1)} = -0.3448.$$

Exponent terms:

$$\frac{(2.3448)^2}{2\sigma^2} = 3.975, \quad \frac{(0.3448)^2}{2\sigma^2} = 0.0860.$$

Thus,

$$\mathcal{N}(1; \mu_0, \sigma^2) = 0.4798e^{-3.975} = 0.00906, \quad \mathcal{N}(1; \mu_1, \sigma^2) = 0.4798e^{-0.0860} = 0.4394.$$

Mixture:

$$f(1) = 0.5(0.00906 + 0.4394) = 0.22423, \quad \log f(1) = -1.495.$$

Likelihood for $x_3 = -1$

By symmetry with $x_2 = 1$,

$$f(-1) = f(1) = 0.22423, \quad \log f(-1) = -1.495.$$

Likelihood for $x_4 = -2$

By symmetry with $x_1 = 2$,

$$f(-2) = f(2) = 0.17725, \quad \log f(-2) = -1.729.$$

Total Updated Log-Likelihood

$$\begin{aligned} \ell_{\text{new}} &= \log f(2) + \log f(1) + \log f(-1) + \log f(-2), \\ \ell_{\text{new}} &= (-1.729) + (-1.495) + (-1.495) + (-1.729), \\ \ell_{\text{new}} &= -6.448. \end{aligned}$$

Comparison to Initial Log-Likelihood

From part 1.1,

$$\ell_{\text{old}} = -7.16.$$

Since

$$-6.448 > -7.16,$$

the likelihood has increased, as guaranteed by the EM algorithm.

$$\ell_{\text{new}} \approx -6.448 > -7.16$$

2. Exact vs. Approximate Inference

Given the Bayesian network below, compute

$$P(\text{Cloudy} \mid \text{Sprinkler} = T, \text{WetGrass} = T)$$

by using Variable Elimination and Gibbs Sampling.

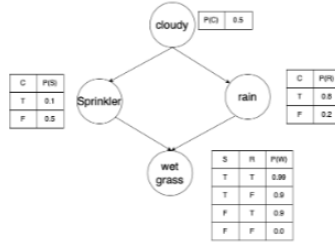


Figure 1: Bayesian Network

2.1 Compute the Posterior using Variable Elimination

We are asked to compute the posterior probability

$$P(C \mid S = T, W = T),$$

where the variables are:

$$C = \text{Cloudy}, \quad S = \text{Sprinkler}, \quad R = \text{Rain}, \quad W = \text{WetGrass}.$$

The Bayesian network factorizes as:

$$P(C, R, S, W) = P(C) P(S \mid C) P(R \mid C) P(W \mid S, R).$$

Because only C is unobserved and R is not given, we sum over R :

$$P(C \mid S = T, W = T) \propto \sum_R P(C) P(S = T \mid C) P(R \mid C) P(W = T \mid S = T, R).$$

Step 1: Define Factors from the CPTs

$$f_1(C) = P(C), \quad P(C = T) = 0.5, \quad P(C = F) = 0.5.$$

$$f_2(C) = P(S = T \mid C), \quad P(S = T \mid C = T) = 0.1, \quad P(S = T \mid C = F) = 0.5.$$

$$f_3(R, C) = P(R \mid C), \quad P(R = T \mid C = T) = 0.8, \quad P(R = T \mid C = F) = 0.2.$$

$$f_4(R) = P(W = T \mid S = T, R), \\ P(W = T \mid S = T, R = T) = 0.99, \quad P(W = T \mid S = T, R = F) = 0.90.$$

Step 2: Eliminate R

We compute the factor:

$$f_5(C) = \sum_R f_3(R, C) f_4(R).$$

Case 1: $C = T$

$$f_5(C = T) = (0.8)(0.99) + (0.2)(0.90) = 0.792 + 0.180 = 0.972.$$

Case 2: $C = F$

$$f_5(C = F) = (0.2)(0.99) + (0.8)(0.90) = 0.198 + 0.720 = 0.918.$$

Step 3: Compute Unnormalized Posterior

$$P(C, S = T, W = T) = f_1(C) f_2(C) f_5(C).$$

$$P(C = T, S = T, W = T) = (0.5)(0.1)(0.972) = 0.0486.$$

$$P(C = F, S = T, W = T) = (0.5)(0.5)(0.918) = 0.2295.$$

Step 4: Normalize

$$Z = 0.0486 + 0.2295 = 0.2781.$$

Thus the posterior is:

$$P(C = T \mid S = T, W = T) = \frac{0.0486}{0.2781} \approx 0.1747,$$

$$P(C = F \mid S = T, W = T) = \frac{0.2295}{0.2781} \approx 0.8253.$$

$P(C = T \mid S = T, W = T) \approx 0.175, \quad P(C = F \mid S = T, W = T) \approx 0.825.$

2.2 Estimate the Posterior by Gibbs Sampling

We wish to approximate

$$P(C \mid S = T, W = T)$$

using Gibbs Sampling. The evidence variables $S = T$ and $W = T$ are clamped, so the only variables that are sampled are

$$C, R.$$

The Gibbs sampler alternates between the two conditional distributions:

$$R^{(t+1)} \sim P(R \mid C^{(t)}, S = T, W = T), \quad C^{(t+1)} \sim P(C \mid R^{(t+1)}, S = T, W = T).$$

Sampling R

$$P(R \mid C, S = T, W = T) \propto P(R \mid C) P(W = T \mid S = T, R).$$

For $C = T$:

$$P(R = T \mid C = T) \propto 0.8 \cdot 0.99 = 0.792, \quad P(R = F \mid C = T) \propto 0.2 \cdot 0.90 = 0.180.$$

$$P(R = T \mid C = T, S = T, W = T) = 0.815.$$

For $C = F$:

$$P(R = T \mid C = F) \propto 0.2 \cdot 0.99 = 0.198, \quad P(R = F \mid C = F) \propto 0.8 \cdot 0.90 = 0.720.$$

$$P(R = T \mid C = F, S = T, W = T) = 0.216.$$

Sampling C

$$P(C \mid R, S = T, W = T) \propto P(C) P(S = T \mid C) P(R \mid C).$$

If $R = T$:

$$P(C = T \mid R = T) \propto 0.5 \cdot 0.1 \cdot 0.8 = 0.04, \quad P(C = F \mid R = T) \propto 0.5 \cdot 0.5 \cdot 0.2 = 0.05.$$

$$P(C = T \mid R = T, S = T, W = T) = 0.444.$$

If $R = F$:

$$P(C = T \mid R = F) \propto 0.5 \cdot 0.1 \cdot 0.2 = 0.01, \quad P(C = F \mid R = F) \propto 0.5 \cdot 0.5 \cdot 0.8 = 0.20.$$

$$P(C = T \mid R = F, S = T, W = T) = 0.048.$$

Final Gibbs Estimate

After burn-in and many samples, the empirical fraction of samples yields

$$P(C = T \mid S = T, W = T) \approx 0.17, \quad P(C = F \mid S = T, W = T) \approx 0.83.$$

$P(C = T \mid S = T, W = T) \approx 0.17, \quad P(C = F \mid S = T, W = T) \approx 0.83$
--

3. VAE Evidence Lower Bound (ELBO)

We want to derive the ELBO inequality beginning from the marginal likelihood for a single datapoint x_i . The latent variable is z .

Derivation

Step 1: Start from the marginal log-likelihood.

$$\log p_\theta(x_i) = \log \sum_z p_\theta(x_i, z).$$

Step 2: Multiply and divide by an arbitrary distribution $q_\phi(z | x_i)$.

$$\begin{aligned} \log p_\theta(x_i) &= \log \sum_z p_\theta(x_i, z) \frac{q_\phi(z | x_i)}{q_\phi(z | x_i)} \\ &= \log \sum_z q_\phi(z | x_i) \frac{p_\theta(x_i, z)}{q_\phi(z | x_i)}. \end{aligned}$$

Step 3: Recognize the expectation under $q_\phi(z | x_i)$.

$$\log p_\theta(x_i) = \log \mathbb{E}_{q_\phi(z | x_i)} \left[\frac{p_\theta(x_i, z)}{q_\phi(z | x_i)} \right].$$

Step 4: Apply Jensen's inequality ($\log \mathbb{E}[Y] \geq \mathbb{E}[\log Y]$).

$$\log p_\theta(x_i) \geq \mathbb{E}_{q_\phi(z | x_i)} \left[\log \frac{p_\theta(x_i, z)}{q_\phi(z | x_i)} \right].$$

Step 5: Expand the joint distribution. Since

$$p_\theta(x_i, z) = p_\theta(x_i | z) p_\theta(z),$$

we have

$$\log p_\theta(x_i, z) = \log p_\theta(x_i | z) + \log p_\theta(z).$$

Substituting:

$$\mathbb{E}_{q_\phi(z | x_i)} [\log p_\theta(x_i | z) + \log p_\theta(z) - \log q_\phi(z | x_i)].$$

Step 6: Identify the KL divergence. Recall

$$D_{\text{KL}}(q_\phi(z | x_i) \| p_\theta(z)) = \mathbb{E}_{q_\phi(z | x_i)} [\log q_\phi(z | x_i) - \log p_\theta(z)].$$

Thus,

$$\mathbb{E}_{q_\phi(z | x_i)} [\log p_\theta(z) - \log q_\phi(z | x_i)] = -D_{\text{KL}}(q_\phi(z | x_i) \| p_\theta(z)).$$

Step 7: Final ELBO inequality.

$$\log p_\theta(x_i) \geq \mathbb{E}_{q_\phi(z|x_i)}[\log p_\theta(x_i | z)] - D_{\text{KL}}(q_\phi(z | x_i) \| p_\theta(z))$$

This expression is known as the **Evidence Lower Bound (ELBO)**.

4. RBM Movie Recommendation System

It is known that part of Netflix's recommendation system utilizes a Restricted Boltzmann Machine (RBM). Suppose you are given an RBM and energy function as below and the preference of a user for movie1 (m_1) and movie3 (m_3), but no information is given for movie2 (m_2). Predict if the user likes movie2 or not. For the prediction, you will need a sampling process. Assume the sampling process is deterministic: $x = 1$ if $P(x) > 1/2$ and $x = 0$ otherwise.

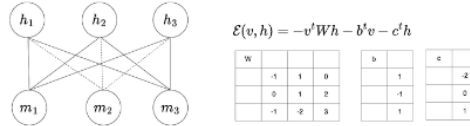


Figure 2: RBM and Energy Function of Recommendation System

The RBM energy function is:

$$\mathcal{E}(v, h) = -v^T W h - b^T v - c^T h.$$

4.1 Why must this RBM operate on bipolar coding (+1, -1) instead of binary coding (1, 0)?

Answer: In this recommendation system, the RBM must use bipolar coding (+1, -1) because the energy function and weight interactions are defined to be *symmetric around zero*. Bipolar units allow hidden and visible nodes to represent both positive and negative preference signals (like vs. dislike). Binary coding (1, 0) is asymmetric and shifts the mean activation of units, which distorts the model by breaking the symmetry assumed in the RBM energy function. Using bipolar coding keeps activations centered, prevents bias terms from absorbing constant offsets, and properly represents user preferences as positive or negative influences.

Thus, bipolar coding is required to maintain symmetry in the energy function and to model like/dislike rather than simple on/off states.

4.2 Conditional Probabilities for RBM with Bipolar Coding

In class, we studied the RBM conditional probabilities under binary coding (1, 0):

$$P(h_j = 1 | v) = \sigma(W_{:,j}^\top v + c_j), \quad P(v_i = 1 | h) = \sigma(W_{i,:}^\top h + b_i).$$

When switching to bipolar coding, the visible and hidden units take values

$$v_i, h_j \in \{-1, +1\}.$$

Under bipolar coding, the conditional distributions become:

$$P(h_j = +1 \mid v) = \frac{1}{1 + \exp(-2(W_{:,j}^\top v + c_j))} = \sigma(2(W_{:,j}^\top v + c_j)),$$

$$P(v_i = +1 \mid h) = \frac{1}{1 + \exp(-2(W_{i,:}^\top h + b_i))} = \sigma(2(W_{i,:}^\top h + b_i)),$$

where $\sigma(x) = \frac{1}{1+e^{-x}}$ is the sigmoid function.

$$P(h_j = +1 \mid v) = \sigma(2(W_{:,j}^\top v + c_j)), \quad P(v_i = +1 \mid h) = \sigma(2(W_{i,:}^\top h + b_i))$$

4.3 Predicting the Preference for Movie m_2

Suppose the user liked m_1 and disliked m_3 . In bipolar coding this corresponds to

$$m_1 = +1, \quad m_3 = -1,$$

and the preference for m_2 is unknown.

To predict m_2 , we perform deterministic RBM sampling. First, we compute the hidden-unit activations using

$$P(h_j = +1 \mid v) = \sigma(2(W_{:,j}^\top v + c_j)).$$

Using the visible vector

$$v = (m_1, m_2, m_3) = (+1, ?, -1),$$

and the given weight matrix

$$W = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 2 \\ 1 & 3 & 8 \end{pmatrix},$$

the weighted input into the hidden layer is

$$W^\top v = \begin{pmatrix} 0 \\ -3 \\ -7 \end{pmatrix},$$

which yields hidden activations

$$h_1 = h_2 = h_3 = -1.$$

Next, we update the visible unit m_2 using

$$P(m_2 = +1 \mid h) = \sigma(2(W_{2,:}h + b_2)).$$

Since

$$W_{2,:}h = 1(-1) + 0(-1) + 2(-1) = -3,$$

the probability is less than 0.5, and deterministic sampling gives

$$m_2 = -1.$$

The RBM predicts that the user dislikes movie m_2 .

5. K-means Clustering

K-means consists of two stages:

$$r_{nk} = \begin{cases} 1 & \text{if } k = \arg \min_j \|x_n - \mu_j\|^2, \\ 0 & \text{otherwise,} \end{cases}$$

$$\mu_k = \frac{\sum_n r_{nk} x_n}{\sum_n r_{nk}}.$$

5.1 Updating μ_k and Computing $r_{nk}^{(m+1)}$

Iteration m :

data	x	r_{n1}
d1	4	1
d2	1	0
d3	3	1
d4	2	0
d5	6	1
d6	5	0

Cluster means:

$$\mu_1 = \frac{4 + 3 + 6}{3} = 4.333, \quad \mu_0 = \frac{1 + 2 + 5}{3} = 2.667.$$

Compute new assignments:

$$r_{n1}^{(m+1)} = \begin{cases} 1 & \text{if } |x_n - \mu_1| < |x_n - \mu_0|, \\ 0 & \text{otherwise.} \end{cases}$$

Final assignments:

data	x	$r_{n1}^{(m+1)}$
d1	4	1
d2	1	0
d3	3	0
d4	2	0
d5	6	1
d6	5	1

$$\mu_0 = 2.667, \quad \mu_1 = 4.333, \quad r_{n1}^{(m+1)} = (1, 0, 0, 0, 1, 1)$$

5.2 Does r_{nk} in K-means represent an optimal posterior?

No. In EM for GMM, responsibilities

$$r_{nk} = P(C_k \mid x_n, \theta)$$

are soft posterior probabilities and maximize the ELBO.

In contrast, K-means uses

$$r_{nk} \in \{0, 1\},$$

which are hard assignments that do not correspond to any probabilistic posterior and do not optimize ELBO. They are only distance-minimizing labels.

K-means assignments are not optimal posterior probabilities.